Name: Class: Date:

Activity 1 · Hands-on · 🏠 At home · ⏱ 4 weeks · 5 min a week

♻️ Compost in a Bottle

You will learn: Decomposers break dead matter into humus and minerals through fragmentation, leaching, catabolism, humification and mineralisation.

YOU NEED

- A large plastic bottle with the top cut off
- Garden soil
- Dry leaves
- Vegetable and fruit peels (no meat or oil)
- Water spray
- A nail to make air holes

⚠️ **Safety:** Ask an adult to make holes. Wash hands after handling soil.

STEPS

1. Make small air holes in the bottle.
2. Add layers: soil, dry leaves, peels, soil, leaves, peels.
3. Spray water to keep it just moist; mark the height of the layers.
4. Each week note the height, colour, smell, and any organisms (worms, fungi, insects).
5. After 4 weeks, compare the bottom layer with fresh soil.

OBSERVATION TABLE

Week	Height (cm)	Colour	Smell	Organisms seen

QUESTIONS

1. Why did the height of the contents decrease?

2. Why were air holes needed?

3. Name the five steps of decomposition.

WHAT IS HAPPENING? — THE SCIENCE

Detritivores like earthworms fragment the litter; water-soluble nutrients leach down; bacteria and fungi break it down by enzymes (catabolism); dark humus forms (humification) and nutrients are released (mineralisation). Warm, moist, aerated conditions speed this up.

Key terms: **decomposition** **detritivores** **humus** **mineralisation**

Activity 2 · Hands-on · 🌳 Outdoors · ⏱ 1 hour

🕸️ Map a Food Web

> **You will learn:** Energy flows through trophic levels; only about 10% passes to the next level.

YOU NEED

- Notebook
- A garden, field or pond to observe for 20 minutes
- Coloured pens

> ⚠️ **Safety:** Stay away from deep water; do not catch animals.

STEPS

1. List every organism you see or know lives there (plants, insects, birds, frogs, worms, fungi).
2. Draw arrows from food to eater ("is eaten by").
3. Mark producers, herbivores, carnivores and decomposers.
4. Trace two food chains and assign trophic levels.
5. If producers trap 10,000 J, calculate the energy at each level using the 10% law.

OBSERVATION TABLE

Organism	Eats	Eaten by	Trophic level	Energy (J)

QUESTIONS

1. Why is energy flow unidirectional?

2. How much energy reaches the 4th trophic level from 10,000 J?

3. Why are there few top carnivores?

WHAT IS HAPPENING? — THE SCIENCE

Food chains link to form a food web. Energy flow is unidirectional, and at each transfer most energy is lost as heat, so only about 10% is passed on (Lindeman's 10 percent law). That is why food chains rarely have more than 4–5 levels.

Key terms: **food web** **trophic level** **10 percent law** **producers**

Digital activity

💻 Energy Pyramid Builder

> Set the sunlight energy captured by producers and the transfer efficiency, and watch the pyramid.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch12/2**

> **You will learn:** Energy at each level ≈ 10% of the level below. The pyramid of energy is always upright.

THINK & WRITE

1. How many levels can be supported when the transfer is 10%? And 20%?

2. Why can the pyramid of energy never be inverted?